

Performance Testing

Date	1 June 2026
Team ID	XXXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Performance Testing

Step 1: Testing Overview

Field	Details
Testing Tool Used	Local Host
Type of Testing	Unit Testing, Performance Testing, Model Testing
Target Module / API	Credit Card Approval Prediction Web Application
Test Environment	Local
Test Date	1 June 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Launch the Credit Card Approval Prediction application	1	1 min	Home page loads successfully.
2	Enter valid applicant details (income, employment, credit history, etc.)	1	1 min	Application data is accepted successfully.
3	Verify the Predict button functionality	1	1 min	Approval prediction is generated successfully.
4	View prediction result	1	1 min	Approval or rejection result is displayed accurately.

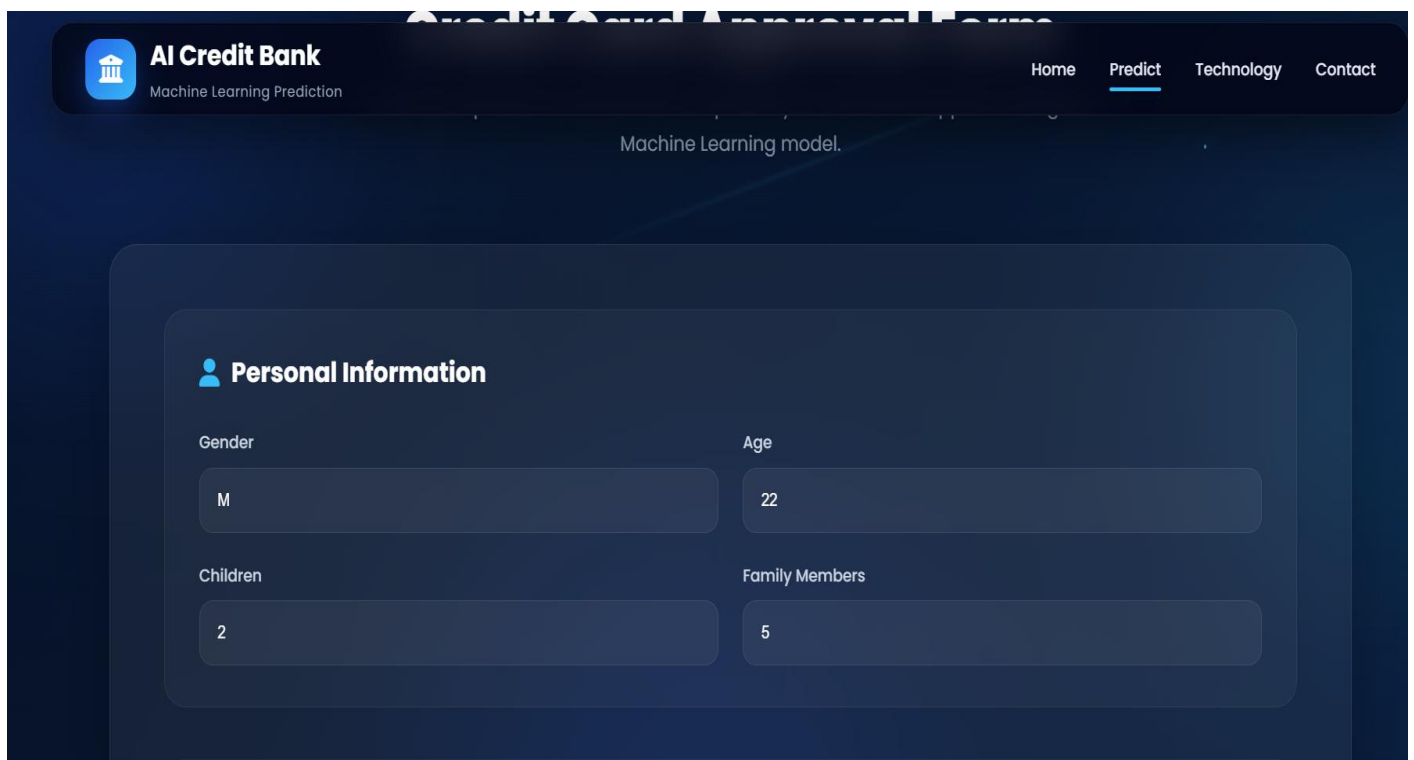
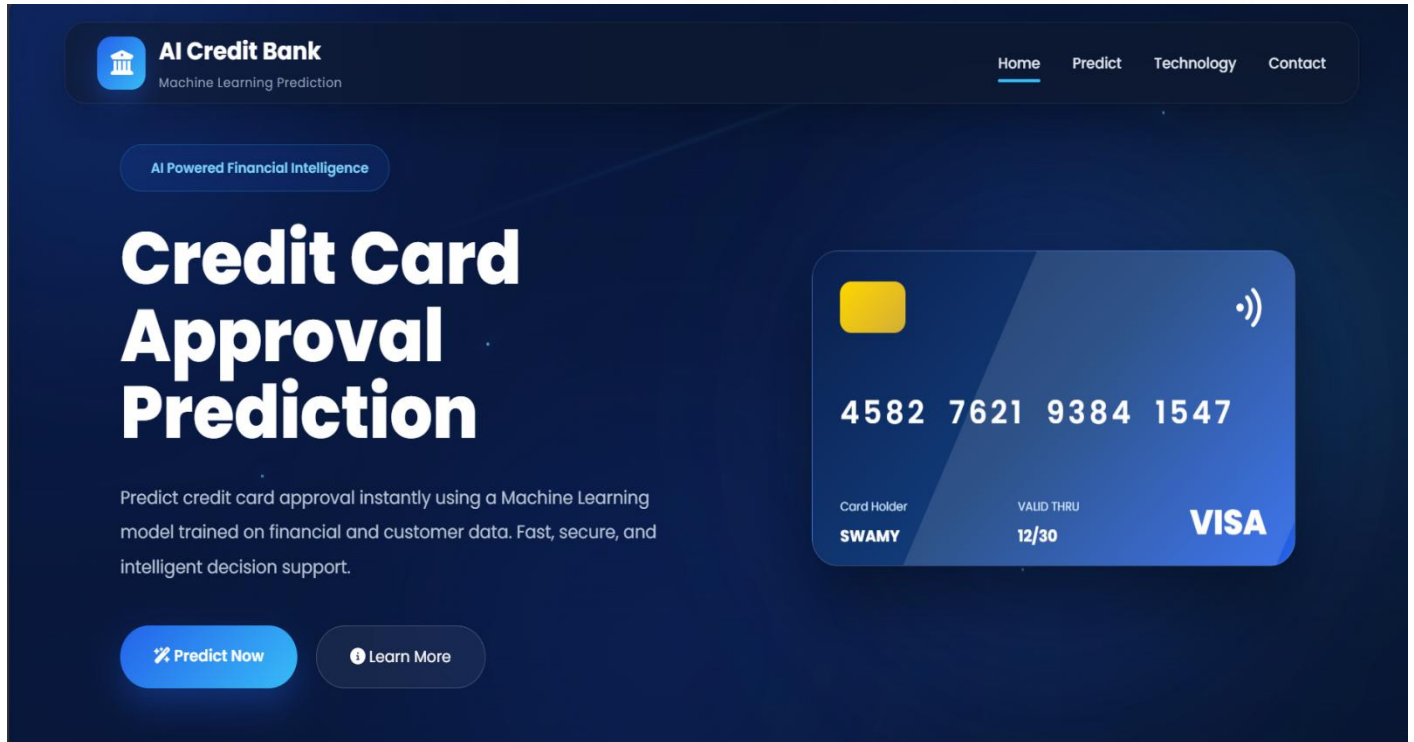
Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	2 sec	Pass	No Issues
2	Response Time (Max)	< 5 seconds	3 sec	Pass	No Issues
3	Throughput (Req/sec)	≥ 1 Req/sec	2 Req/sec	Pass	Stable
4	Error Rate	< 1%	0%	Pass	No Errors
5	CPU Utilization	< 80%	< 65%	Pass	Normal
6	Memory Utilization	< 80%	< 60%	Pass	Normal

Step 4: Observations & Analysis

- The application loads successfully without errors.
- Applicant data is processed correctly and validated before prediction.
- The machine learning model generates approval predictions quickly.
- Frontend, backend, and model integration function smoothly.
- Overall system performance is stable with low resource utilization.

Step 5: Screenshots / Evidence



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**AI Credit Bank**
Machine Learning Prediction

Home Predict Technology Contact

Annual Income

250000

Income Type

State servant

Education

Academic degree

Occupation

Managers

 **Property Details**

Own Car

Y

Own House

Y

Housing Type

House / apartment

Family Status

Married

**AI Credit Bank**
Machine Learning Prediction

Home Result

 AI Prediction Completed



Credit Card Approved
Our Machine Learning model has analyzed the submitted financial information and generated the following prediction.
Prediction Confidence 70.8%

 **AI Risk Analysis**



Financial Stability

High



Credit Risk

Low

EXPLORER

CREDITCARD

dataset

application_record.csv

credit_record.csv

final_dataset.csv

model

credit_card_model.pkl

encoders.pkl

features.pkl

form_options.pkl

static

css

style.css

images

credit-card.png

js

script.js

templates

index.html

result.html

venv

.gitignore

app.py

README.md

requirements.txt

train.py

train.py

app.py

.gitignore

requirements.txt

index.html

result.html

style.css

app.py

1 # =====

2 # Credit Card Approval Prediction System

3 # app.py (Part 1)

4 # =====

5

6 from flask import Flask, render_template, request

7 import pandas as pd

8 import joblib

9 import numpy as np

10

11 # =====

12 # LOAD FILES

13 # =====

14

15 model = joblib.load("model/credit_card_model.pkl")

16 encoders = joblib.load("model/encoders.pkl")

17 features = joblib.load("model/features.pkl")

18 form_options = joblib.load("model/form_options.pkl")

19

20 # =====

21 # CREATE FLASK APP

22 # =====

23

24 app = Flask(__name__)

25

26 # =====

27 # HOME PAGE

28 # =====

29

30 @app.route("/")

31 def home():

32

33 return render_template(

34 "index.html",

35 form_options=form_options

36)

37

38 # =====

39 # PREDICTION

40 # =====

41

42 @app.route("/predict", methods=["POST"])

43 def predict():

44

45 data = {}